

Task 4: Mini SQL Agent (Agentic System Task)

Goal

Build a simple **agentic system that thinks, acts, and corrects itself**

Instructions

Create a FastAPI endpoint:

POST /agent/sql

Input

```
{  
  "question": "How many shipped orders are from USA customers?"  
}
```

Required Agent Flow

Your system must follow this process:

Step 1: Understand Query

- Identify intent
 - Detect tables and columns
-

Step 2: Generate SQL

- Write SQL query based on structure
-

Step 3: Execute Query

- Run SQL in PostgreSQL
-

Step 4: Error Handling (IMPORTANT)

If query fails:

- Read error message
 - Fix query
 - Retry up to **3 times maximum**
-

Step 5: Final Output

- Return SQL result
 - Convert result into natural language summary
-

Expected Output

```
{  
  "sql": "SELECT COUNT(*) FROM orders o JOIN customers c ...",  
  "result": 42,  
  "summary": "There are 42 shipped orders from customers in USA.",  
  "status": "success"  
}
```

Rules

- Use only SELECT queries
 - Must implement retry logic (max 3 attempts)
 - Must log:
 - decomposition step
 - SQL generation
 - execution time
 - Must use safe query validation
 - Must include fallback response if all retries fail
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What You Will Learn

- How AI agents work internally
 - How systems self-correct using feedback
 - How real-world AI APIs are built
-

Evaluation Criteria

You will be assessed on:

Understanding

- Correct decomposition of queries

Implementation

- Working SQL generation
- Correct API structure

Agent Behavior

- Retry mechanism
- Error handling

System Design

- Clean FastAPI structure
- Logging implementation
- Safe SQL execution

Performance on the test dataset:

https://docs.google.com/spreadsheets/d/1lgh-Wk6wJMGSEZiQh_ILqVpuRn4UbPWFUnYzB1RZ9Qc/edit?gid=1012683343#gid=1012683343

How well your system work against these queries.

Final Outcome

By the end of this assignment, you will be able to build:

A working **AI-powered SQL agent system** that mimics real production-level data assistants like analytics chatbots and BI tools.